

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (EC) Examination****APRIL-MAY 2018****EC 209 Microprocessors & Microcontroller****Date: 02.05.2018, Wednesday****Time: 10:00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION –I

- Q-1 Do as Directed. 07**
- a.** When the micro controller executes some arithmetic operations, then the flag bits of which register are affected? **01**
 - a) PSW b) SP
 - c) DPTR d) PC
 - b.** How many bytes of bit addressable memory is present in 8051 based micro controllers? **01**
 - a) 8bytes b) 32 bytes
 - c) 16 bytes d) 128 bytes
 - c.** What is the meaning of the instruction MOV A,05H ? **01**
 - a) data 05H is stored in the accumulator
 - b) fifth bit of accumulator is set to one
 - c) address 05H is stored in the accumulator
 - d) none of the mentioned
 - d.** To initialize any port of 8051 as an output port what value is to be given to it? **01**
 - a) 0xFF
 - b) 0x00
 - c) 0x01
 - d) A port is by default an output port
 - e.** Which out of the four ports of 8051 needs a pull-up resistor for using it as an input or an output port? **01**
 - a) PORT 0
 - b) PORT 1
 - c) PORT 2

d) PORT 3

f. Which instruction is used to check the status of a single bit? **01**

a) MOV A,P0

b) ADD A,#05H

c) JNB P0.0, label

d) CLR P0.05H

g. What is the frequency of the clock that is being used as the clock source for the timer? **01**

a) some externally applied frequency f

b) controller's crystal frequency f

c) controller's crystal frequency /12

d) externally applied frequency/12

Q-2 (a) Write an assembly language program to generate 1KHz square wave on P1.1 using Timer 0.(Assume Crystal Frequency=12MHz). **07**

(b) Write an assembly language program to send "SUMMER 2018" Serially with baud rate of 4800 using serial communication. **07**

OR

(b) Explain data exchange operation between external memory and 8051 with suitable examples. **07**

Q-3 (a) Interface 2 kb RAM and 1kb ROM with 8085 processor to satisfy the total memory requirement of 4 kb RAM(starting address is 0000h) and 2 kb ROM(starting address is 8000h). **07**

(b) Differentiate between Microprocessor and Microcontroller. **04**

OR

(b) Explain internal memory organization of 8051 with neat and clean diagram. **04**

(c) Explain functionality of following pin of 8085. **03**

(i) ALE (ii) HOLD (iii) READY

SECTION – II

Q-4 Interface 8255 programmable peripheral interface with 8085.(Assume PORTA address of 8255 is B0h).Write a control word to initialize PORTC as input and PORTA and PORTB as output in Mode 1. **07**

- Q-5(a)** List SFRs available in 8051 and Explain TMOD register in detail . **07**
- (b)** Interface 8254 programmable timer/counter with 8085 and Write an assembly language code to generate 10 KHz square wave on counter 0(Assume input clock Frequency to 8254 is 2 MHz). **07**

OR

- (b)** Write an assembly language program to find minimum number from an array of 10 bytes located in to internal memory of 8051. **07**
- Q-6(a)** Interface 4kb of ROM with 8051 with neat and clean diagram (address range 0000H to 0FFFH). **07**
- (b)** Explain SCON register in detail. **07**
